# PINNsFormer for the 4D Hodgkin-Huxley Neuron Model: A Comprehensive Literature Review, Technical Analysis, and Research Proposal

## Part 1: Neuroscience Foundations

The electrical excitability of neuronal membranes serves as the primary mechanism for signal transmission, integration, and processing within the central and peripheral nervous systems. At the cellular level, this excitability is governed by the trans-membrane potential, which is established by the unequal distribution of ions across the lipid bilayer and the selective permeability of voltage-gated ion channels. In 1952, Alan Hodgkin and Andrew Huxley published a series of pioneering papers that mathematically characterized these biophysical phenomena. Using voltage-clamp experiments on the giant axon of the longfin inshore squid, they isolated the individual ionic currents that generate the action potential, establishing the parallel-conductance model as the foundation of modern computational neuroscience.

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                 Extracellular Medium (Reference Potential = 0 mV)
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       |                     |                      |
|
       |                     |                      |
|
     +---                  +----+                 +----+
+----+
     |   |                 |   |                  |   |
|   |
     | C |                 |g_Na                  | g_K
| g_L
     | m |                 |   |                  |   |
|   |
     +---                  +----+                 +----+
+----+
       |                     |                      |
|
       |                   - | +                  - | +
- | +
       |                  ( E_Na )               (  E_K )
(  E_L )
       |                     |                      |
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       |                     |                      |
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-----------------------------------------------------------------------
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                 Intracellular Medium (Membrane Potential = V)
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The neuronal membrane functions as a capacitor, C_m, which separates the electrical charges of the intracellular and extracellular electrolytes. Active trans-membrane currents flow through specialized macromolecular protein structures known as voltage-gated ion channels, which exhibit selective permeability to sodium (\text{Na}^+) and potassium (\text{K}^+) ions. Additionally, a passive background leakage path, primarily carrying chloride (\text{Cl}^-) and other secondary ions, maintains the membrane's resting potential.
An action potential is generated through a highly regulated sequence of electrochemical events :

1. **Resting Phase**: The resting membrane potential stabilizes near -65\text{ mV}, primarily driven by passive leak conductances and the active transport of ions by the \text{Na}^+/\text{K}^+-ATPase pump, which exchanges three intracellular \text{Na}^+ ions for two extracellular \text{K}^+ ions.
2. **Depolarization Threshold**: Upon the injection of an external depolarizing stimulus, the membrane potential rises. If the potential exceeds a critical threshold (typically between -55\text{ mV} and -50\text{ mV}), the activation gates of the voltage-gated sodium channels rapidly open.
3. **Sodium Influx (Upstroke)**: The opening of sodium channels drives a rapid inward flow of \text{Na}^+ ions down their concentration and electrical gradients. This positive feedback loop—where depolarization increases sodium conductance, causing further depolarization—drives the membrane potential toward the sodium reversal potential (E_{\text{Na}} \approx +50\text{ mV} to +115\text{ mV}).
4. **Repolarization**: Near the peak of the action potential, two slower, voltage-dependent processes occur. First, the sodium channel inactivation gates close, halting the inward sodium current. Second, the voltage-gated potassium channels open, driving an outward flow of \text{K}^+ ions that repolarizes the membrane back toward its negative resting state.
5. **Hyperpolarization and Recovery**: Because the voltage-gated potassium channels close slowly, the sustained efflux of \text{K}^+ ions temporarily hyperpolarizes the membrane potential below its resting value. The membrane potential gradually returns to the resting baseline through passive leak channels and active pump mechanisms.

The Hodgkin-Huxley model is a foundational dynamical system in computational neuroscience. It provides a biophysical explanation for key neuronal properties, including excitability (the threshold-dependent generation of large-amplitude voltage excursions), spike generation, and refractory periods. The absolute refractory period, caused by the inactivation of sodium channels, prevents back-propagation and limits the maximum firing frequency of the neuron. The relative refractory period, caused by sustained potassium conductance, increases the threshold for subsequent action potentials.
These properties govern neural coding, which is the language the brain uses to represent and transmit information. Computational models leverage these biophysical dynamics to investigate rate coding (where information is encoded in the frequency of action potentials) and temporal coding (where information is carried by the precise timing of individual spikes), making the Hodgkin-Huxley model central to understanding neural network computation and

neuromodulation.

# Part 2: Full Mathematical Formulation

The 4D Hodgkin-Huxley model represents the electrical state of a space-clamped patch of membrane using a system of four coupled, non-linear ordinary differential equations. Applying Kirchhoff's current law to the equivalent circuit, the total trans-membrane current density is the sum of the capacitive current and the individual ionic currents :

$$C_m \frac{dV}{dt} = I_{\text{ext}}(t) - I_{\text{Na}}(V, m, h) - I_{\text{K}}(V, n) - I_{\text{L}}(V)$$

Substituting the voltage-dependent conductances and ionic driving forces, the complete membrane potential equation is formulated as :

$$C_m \frac{dV}{dt} = I_{\text{ext}}(t) - \bar{g}_{\text{Na}}m^3h(V - E_{\text{Na}}) - \bar{g}_{\text{K}}n^4(V - E_{\text{K}}) - g_{\text{L}}(V - E_{\text{L}})$$

The three gating variables $m(t)$ (sodium activation), $h(t)$ (sodium inactivation), and $n(t)$ (potassium activation) represent the probability of individual channel subunits being in their open states. These variables are bounded on the interval $$ and evolve according to first-order kinetic rate equations :

$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m \quad \frac{dh}{dt} = \alpha_h(V)(1 - h) - \beta_h(V)h \quad \frac{dn}{dt} = \alpha_n(V)(1 - n) - \beta_n(V)n$$

The transition rates $\alpha_i(V)$ and $\beta_i(V)$ represent the voltage-dependent opening and closing rates of the gating subunits, respectively. The classical parameterization developed by Hodgkin and Huxley for the squid giant axon at a temperature of $6.3^\circ\text{C}$ is given by the following empirical formulations :

$$\alpha_m(V) = \frac{0.1(V + 40)}{1 - \exp\left(-\frac{V + 40}{10}\right)} \quad \beta_m(V) = 4\exp\left(-\frac{V + 65}{18}\right) \quad \alpha_h(V) = 0.07\exp\left(-\frac{V + 65}{20}\right) \quad \beta_h(V) = \frac{1}{1 + \exp\left(-\frac{V + 35}{10}\right)} \quad \alpha_n(V) = \frac{0.01(V + 55)}{1 - \exp\left(-\frac{V + 55}{10}\right)} \quad \beta_n(V) = 0.125\exp\left(-\frac{V + 65}{80}\right)$$

*Note: In these formulations, $V$ represents the membrane potential in millivolts ($\text{mV}$), relative to a resting potential normalized to $-65\text{ mV}$.*

To analyze the system mathematically, let the state vector be $\mathbf{x}(t) = [V(t), m(t), h(t), n(t)]^T \in \mathbb{R}^4$. The 4D Hodgkin-Huxley model can be rewritten as a autonomous, non-linear dynamical system:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}(t); \boldsymbol{\theta})$$

where $\boldsymbol{\theta}$ denotes the vector of biophysical parameters, and the vector field $\mathbf{f}: \mathbb{R}^4 \to \mathbb{R}^4$ is defined explicitly as:

$$\mathbf{f}(\mathbf{x}(t); \boldsymbol{\theta}) = \begin{bmatrix} \frac{1}{C_m} \left( I_{\text{ext}}(t) - \bar{g}_{\text{Na}}x_2^3x_3(x_1 - E_{\text{Na}}) - \bar{g}_{\text{K}}x_4^4(x_1 - E_{\text{K}}) - g_{\text{L}}(x_1 - E_{\text{L}}) \right) \\ \alpha_m(x_1)(1 - x_2) - \beta_m(x_1)x_2 \\ \alpha_h(x_1)(1 - x_3) - \beta_h(x_1)x_3 \\ \alpha_n(x_1)(1 - x_4) - \beta_n(x_1)x_4 \end{bmatrix}$$

| Gating Term / Parameter | Physical Meaning | Standard Biophysical Value | Units |
|---|---|---|---|
| $V$ | Trans-membrane electrical potential | -65.0 (at rest) | $\text{mV}$ |
| $m$ | Sodium channel activation probability | 0.0529 (at rest) | Dimensionless |
| $h$ | Sodium channel | 0.5961 (at rest) | Dimensionless |

| Gating Term / Parameter | Physical Meaning | Standard Biophysical Value | Units |
|---|---|---|---|
| | inactivation probability | | |
| $n$ | Potassium channel activation probability | 0.3177 (at rest) | Dimensionless |
| $C_m$ | Specific membrane capacitance | 1.0 | $\mu\text{F/cm}^2$ |
| $\bar{g}_{\text{Na}}$ | Max sodium channel conductance | 120.0 | $\text{mS/cm}^2$ |
| $\bar{g}_{\text{K}}$ | Max potassium channel conductance | 36.0 | $\text{mS/cm}^2$ |
| $g_{\text{L}}$ | Background leakage conductance | 0.3 | $\text{mS/cm}^2$ |
| $E_{\text{Na}}$ | Equilibrium potential for sodium ions | +50.0 | $\text{mV}$ |
| $E_{\text{K}}$ | Equilibrium potential for potassium ions | -77.0 | $\text{mV}$ |
| $E_{\text{L}}$ | Equilibrium potential for leak ions | -54.4 | $\text{mV}$ |
| $I_{\text{ext}}$ | Externally injected stimulation current | Variable (typically 0.0 - 20.0) | $\mu\text{A/cm}^2$ |

The structural complexity of this vector field stems from several coupled non-linearities. First, the voltage equation contains multiplicative terms between the state variables ($x_2^3x_3$ and $x_4^4$) and linear voltage differences, representing non-linear, voltage-dependent conductances. Second, the gating equations are coupled back to the voltage equation through the highly non-linear, transcendental rate functions $\alpha_i(V)$ and $\beta_i(V)$, creating a feedback loop.

# Part 3: Dynamical Systems Analysis

A dynamical systems analysis of the 4D Hodgkin-Huxley model begins by locating the equilibrium points $\mathbf{x}^* = [V^*, m^*, h^*, n^*]^T$ where the vector field vanishes :
$$\mathbf{f}(\mathbf{x}^*; \boldsymbol{\theta}) = \mathbf{0}$$
Since m, h, n are in algebraic steady-states at any fixed voltage, their equilibrium values are determined directly by the voltage coordinate :
$$w^*(V^*) = w_{\infty}(V^*) = \frac{\alpha_w(V^*)}{\alpha_w(V^*) + \beta_w(V^*)}, \quad w \in \{m, n, h\}$$
Substituting these algebraic relations into the voltage equation yields a transcendental root-finding problem for $V^*$:
$$I_{\text{ext}} - \bar{g}_{\text{Na}}\left[m_{\infty}(V^*)\right]^3h_{\infty}(V^*)(V^* - E_{\text{Na}}) - \bar{g}_{\text{K}}\left[n_{\infty}(V^*)\right]^4(V^* - E_{\text{K}}) - g_{\text{L}}(V^* - E_{\text{L}}) = 0$$
For a constant external current $I_{\text{ext}}$, this equation typically has a unique real root representing a single equilibrium point, which corresponds to the resting state at low currents.
To assess the stability of $\mathbf{x}^*$, we compute the Jacobian matrix $\mathbf{J}(\mathbf{x}) = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}$ :
\mathbf{J}(\mathbf{x}) = \begin{bmatrix} J_{11} & J_{12} & J_{13} & J_{14} \\ J_{21} & J_{22} & 0

$$& 0 \\ J_{31} & 0 & J_{33} & 0 \\ J_{41} & 0 & 0 & J_{44} \end{bmatrix}$$

The individual partial derivatives are formulated as follows :

$$J_{11} = \frac{\partial f_1}{\partial V} = -\frac{1}{C_m} \left( \bar{g}_{\text{Na}}m^3h + \bar{g}_{\text{K}}n^4 + g_{\text{L}} \right) J_{12} = \frac{\partial f_1}{\partial m} = -\frac{3}{C_m} \bar{g}_{\text{Na}}m^2h(V - E_{\text{Na}}) J_{13} = \frac{\partial f_1}{\partial h} = -\frac{1}{C_m} \bar{g}_{\text{Na}}m^3(V - E_{\text{Na}}) J_{14} = \frac{\partial f_1}{\partial n} = -\frac{4}{C_m} \bar{g}_{\text{K}}n^3(V - E_{\text{K}})$$

For $w \in \{m, h, n\}$, the gating-associated derivatives are:

$$J_{w,1} = \frac{\partial f_i}{\partial V} = \alpha_w'(V)(1 - w) - \beta_w'(V)w, \quad \text{for } i \in \{2, 3, 4\} J_{w,w} = \frac{\partial f_i}{\partial w} = -(\alpha_w(V) + \beta_w(V)) = -\frac{1}{\tau_w(V)}$$

The stability of the system is governed by the eigenvalues $\lambda \in \mathbb{C}$ of the characteristic equation $\det(\mathbf{J}(\mathbf{x}^*) - \lambda \mathbf{I}) = 0$. At $I_{\text{ext}} = 0$, all eigenvalues have negative real parts, indicating that the rest state is a stable focus. As $I_{\text{ext}}$ is increased, the steady-state voltage depolarizes. At a critical bifurcation parameter $I_{\text{ext}} \approx 9.78\,\mu\text{A/cm}^2$, a pair of complex-conjugate eigenvalues crosses the imaginary axis from the left to the right half-plane :

$$\text{Re}(\lambda_{1,2}) = 0, \quad \text{Im}(\lambda_{1,2}) \neq 0$$

This transition corresponds to a **subcritical Andronov-Hopf bifurcation**. Because the bifurcation is subcritical, an unstable limit cycle coexists with the stable focus for a range of currents below the bifurcation point ($6.26\,\mu\text{A/cm}^2 < I_{\text{ext}} < 9.78\,\mu\text{A/cm}^2$). At $I_{\text{ext}} \approx 6.26\,\mu\text{A/cm}^2$, the unstable limit cycle collides with a stable limit cycle in a saddle-node bifurcation of limit cycles, creating a bistable regime where the stable resting focus and stable limit cycle (tonic spiking) coexist.

```
                   Stable
Rep[span_90](start_span)[span_90](end_span)etitive Spiking (Limit
Cycle)
                  ========================================>
                 /
                / (Saddle-Node of Limit Cycles at I ≈ 6.26)
               /
            ........... (Unstable Limit Cycle)
           .                     \
          .                       \  (Subcritical Hopf at I ≈ 9.78)
--------o-------------------------*-------------------->
Stable Rest State (Focus)        Unstable Equilibrium
--------------------------------->
                                 (Unstable Focus / Source)
```

In phase space, these transitions unfold along low-dimensional manifolds. Because the eigenvalues associated with the gating variables m and h have disparate decay rates, the fast dynamics relax quickly onto a two-dimensional center manifold defined by the slow variables n and h. When $I_{\text{ext}} > 9.78\,\mu\text{A/cm}^2$, the stable focus is lost, and the state trajectory is attracted to a stable limit cycle, tracing out a continuous closed orbit in the 4D state-space. This geometric orbit represents a continuous spike train, where the voltage and gating variables oscillate periodically.

# Part 4: Time Scale Analysis

The 4D Hodgkin-Huxley model is characterized by a high degree of timescale separation. To analyze these timescales, we rewrite the gating equations in the standard form :

$$\frac{dw}{dt} = \frac{w_{\infty}(V) - w}{\tau_w(V)}, \quad w \in \{m, h, n\}$$

where the voltage-dependent activation and inactivation timescales are defined as $\tau_w(V) = \left[\alpha_w(V) + \beta_w(V)\right]^{-1}$. The membrane potential equation also possesses an intrinsic electromagnetic timescale determined by the membrane capacitance and instantaneous conductance :

$$\tau_V(V, m, h, n) = \frac{C_m}{\bar{g}_{\text{Na}}m^3h + \bar{g}_{\text{K}}n^4 + g_{\text{L}}}$$

| Gating Variable | Biophysical Role | Rest Timescale ($\tau$ at rest) | Peak Spiking Timescale ($\tau$ at peak) | Dynamical Class |
|---|---|---|---|---|
| m | Sodium Activation | $\approx 0.1 - 0.5\text{ ms}$ | $\approx 0.05\text{ ms}$ | Ultrafast (Activator) |
| V | Membrane Potential | $\approx 1.0 - 3.0\text{ ms}$ | $\approx 0.1\text{ ms}$ | Fast (State Variable) |
| n | Potassium Activation | $\approx 5.0 - 8.0\text{ ms}$ | $\approx 1.5\text{ ms}$ | Slow (Recovery) |
| h | Sodium Inactivation | $\approx 8.0 - 12.0\text{ ms}$ | $\approx 2.0\text{ ms}$ | Slow (Inactivating) |

The massive separation between the ultrafast sodium activation timescale ($\tau_m \approx 0.1\text{ ms}$) and the slow inactivation and potassium kinetics ($\tau_h, \tau_n \approx 10\text{ ms}$) defines the fast-slow structure of the system. Under nondimensionalization, this structure can be written as:

$$\frac{dV}{dt} = f(V, m, h, n) \epsilon_m \frac{dm}{dt} = g_m(V, m) \frac{dh}{dt} = g_h(V, h) \frac{dn}{dt} = g_n(V, n)$$

where $\epsilon_m = \tau_m / \tau_n \ll 1$ acts as a singular perturbation parameter.

This extreme timescale separation introduces severe mathematical stiffness. Stiffness is measured by the ratio of the eigenvalues of the system Jacobian :

$$S_{\text{ratio}} = \frac{\max_j |\text{Re}(\lambda_j(t))|}{\min_j |\text{Re}(\lambda_j(t))|}$$

In the Hodgkin-Huxley system, stiffness is not static; rather, it is a highly dynamic property that changes during different phases of the action potential :

1. **Resting Phase**: The system exhibits **mild stiffness** ($S_{\text{ratio}} \approx 10 - 50$). All variables relax slowly back to the rest state, and the Jacobian is well-conditioned.
2. **Action Potential Upstroke**: The activation of sodium channels triggers a positive feedback loop. This causes m and V to evolve on timescales that are orders of magnitude faster than h and n. Consequently, the system transitions into a regime of **spike-induced stiffness**, with the stiffness ratio spiking to extreme values ($S_{\text{ratio}} > 10^4 - 10^6$). This transient stiffness causes standard explicit numerical integrators to fail unless they use extremely small, computationally expensive steps ($\Delta t < 1\,\mu\text{s}$).

```
Stiffness Ratio (S_ratio)
  ^
  |          /\  <- Spike-Induced Stiffness (Upstroke Peak, S_ratio >
10^5)
  |         /  \
  |        /    \
```

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  |       /      \
  |______/        \______  <- Mild Stiffness (Resting Focus /
Undershoot, S_ratio ≈ 10-50)
  +------------------------> Time (t)
```

Compared to standard stiff ODE benchmarks, the Hodgkin-Huxley model presents unique optimization challenges. While the chemical kinetics of the Robertson problem exhibit monotonic stiff decay over long times, and the Van der Pol oscillator exhibits regular relaxation oscillations, the Hodgkin-Huxley model combines highly non-linear, transient spike-induced stiffness with subcritical bifurcation thresholds. This makes it difficult for traditional physics-informed neural networks (PINNs) to resolve.

# Part 5: Classical Physics-Informed Neural Network (PINN) Formulation

To construct a classical PINN for the Hodgkin-Huxley system, the continuous physical state vector $\mathbf{u}(t) = [V(t), m(t), h(t), n(t)]^T$ is parameterized by a fully connected, feedforward neural network $\hat{\mathbf{u}}_\theta(t) = [\hat{V}_\theta(t), \hat{m}_\theta(t), \hat{h}_\theta(t), \hat{n}_\theta(t)]^T$ :

$$\hat{\mathbf{u}}_\theta(t) = \text{MLP}(t; \theta)$$

where $t \in$ is the single coordinate input, and $\theta$ represents the network weights and biases. Through automatic differentiation (AD) using the chain rule, temporal derivatives of the state predictions are evaluated exactly without grid discretization errors. This allows the continuous evaluation of physical residuals across the domain :

$$\mathcal{R}_V(\theta; t) = C_m \frac{d\hat{V}_\theta}{dt} - I_{\text{ext}}(t) + \bar{g}_{\text{Na}}\hat{m}_\theta^3\hat{h}_\theta(\hat{V}_\theta - E_{\text{Na}}) + \bar{g}_{\text{K}}\hat{n}_\theta^4(\hat{V}_\theta - E_{\text{K}}) + g_{\text{L}}(\hat{V}_\theta - E_{\text{L}}) \mathcal{R}_m(\theta; t) = \frac{d\hat{m}_\theta}{dt} - \alpha_m(\hat{V}_\theta)(1 - \hat{m}_\theta) + \beta_m(\hat{V}_\theta)\hat{m}_\theta \mathcal{R}_h(\theta; t) = \frac{d\hat{h}_\theta}{dt} - \alpha_h(\hat{V}_\theta)(1 - \hat{h}_\theta) + \beta_h(\hat{V}_\theta)\hat{h}_\theta \mathcal{R}_n(\theta; t) = \frac{d\hat{n}_\theta}{dt} - \alpha_n(\hat{V}_\theta)(1 - \hat{n}_\theta) + \beta_n(\hat{V}_\theta)\hat{n}_\theta$$

The objective function used to optimize the network parameters $\theta$ is formulated as a composite, multi-objective loss function evaluated over collocation points $\mathcal{T}_r$ of size $N_r$, initial conditions $\mathcal{T}_{\text{ic}}$ of size 1 (at $t=0$), and data points $\mathcal{T}_{\text{data}}$ of size $N_{\text{data}}$ :

$$\mathcal{L}_{\text{total}}(\theta) = \lambda_V \mathcal{L}_{\mathcal{R}_V}(\theta) + \lambda_m \mathcal{L}_{\mathcal{R}_m}(\theta) + \lambda_h \mathcal{L}_{\mathcal{R}_h}(\theta) + \lambda_n \mathcal{L}_{\mathcal{R}_n}(\theta) + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}}(\theta) + \lambda_{\text{data}} \mathcal{L}_{\text{data}}(\theta)$$

where the individual loss terms are defined as:

$$\mathcal{L}_{\mathcal{R}_V}(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} \left\| \mathcal{R}_V(\theta; t_i^r) \right\|^2 \mathcal{L}_{\mathcal{R}_w}(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} \left\| \mathcal{R}_w(\theta; t_i^r) \right\|^2, \quad w \in \{m, h, n\} \mathcal{L}_{\text{ic}}(\theta) = \left\| \hat{V}_\theta(0) - V_0 \right\|^2 + \left\| \hat{m}_\theta(0) - m_0 \right\|^2 + \left\| \hat{h}_\theta(0) - h_0 \right\|^2 + \left\| \hat{n}_\theta(0) - n_0 \right\|^2 \mathcal{L}_{\text{data}}(\theta) = \frac{1}{N_{\text{data}}} \sum_{j=1}^{N_{\text{data}}} \left\|$$

$\mathbf{M} \hat{\mathbf{u}}_\theta(t_j) - \mathbf{y}_j \right\|^2$
Here, $\mathbf{y}_j$ represents physical measurements (typically sparse membrane potential recordings), and $\mathbf{M}$ is a projection matrix that maps the 4D state vector to the observable dimensions (e.g., $\mathbf{M} = $ under partial observability of voltage).
Classical, coordinate-based MLP PINNs consistently fail when applied to the Hodgkin-Huxley system near action potential spikes. These training failures are driven by the following mathematical and optimization pathologies:

- **Spectral Bias**: The training dynamics of an MLP, analyzed in the infinite-width limit through the Neural Tangent Kernel (NTK) $\mathbf{K}(\mathbf{z}, \mathbf{z}')$, favor the learning of low-frequency components. The eigenvalues of the NTK decay rapidly as a function of the frequency index: $\sigma_k \propto k^{-2d}$. Near a spike, the voltage signal behaves as a high-frequency localized shock. Standard backpropagation updates fail to make progress along these high-frequency directions, causing the network to produce an over-smoothed resting state that satisfies the passive leak equations but fails to initiate the action potential.
- **Gradient Pathologies**: Computing the backpropagation path for $\nabla_\theta \mathcal{L}_r$ requires taking derivatives through the highly non-linear rate equations $\alpha(V)$ and $\beta(V)$. This introduces products of the MLP weight matrices raised to the power of the derivative order. Near the upstroke, this causes extreme scaling differences: $\|\nabla_\theta \mathcal{L}_r\|_2 \gg \|\nabla_\theta \mathcal{L}_{\text{ic}}\|_2$. The optimizer prioritizes satisfying the local ODE residuals at the cost of violating the temporal causality of the initial condition, trapping the model in a trivial, non-spiking state.

# Part 6: PINNsFormer Architecture Design

To overcome the temporal causality violations and spatial-temporal propagation failures of pointwise MLP-based PINNs, the PINNsFormer architecture transforms coordinate-based physics-informed learning into a sequential learning problem.

```
                 +---------------------------------------+
                 |      Temporal Coordinate (t)          |
                 +-------------------+-------------------+
                                     |

                                     |
                                     v
+---------------------------------------------------------------------+
| Sequence of k coordinates: {t, t + dt, t + 2dt,..., t + (k-1)dt} |
+----------------------------------+----------------------------------+
                                     |

                                     |
                                     v
+---------------------------------------------------------------------+
| Sequence of k latent embeddings in d_model dimension               |
+----------------------------------+----------------------------------+
                                     |
```

```
                  (Self & Cross-Attention over Sequence)
                                  |
                                  v
+------------------------------------------------------------------+
| Latent Sequence after Multi-Head Coordinate Coupling               |
+---------------------------------+--------------------------------+
                                  |

                                  |
                                  v
+------------------------------------------------------------------+
| Dual Sine-Cosine Learnable Wavelet Bases                           |
+---------------------------------+--------------------------------+
                                  |
                      [Linear Projection Head]
                                  |
                                  v
+------------------------------------------------------------------+
| Predicted Sequence of States: {\hat{u}_0, \hat{u}_1,...,
\hat{u}_{k-1}} |
| *Note: \hat{u}_0 is exactly mapped to state prediction \hat{u}(t)  |
+------------------------------------------------------------------+
```

Instead of evaluating the neural network at isolated coordinates, the PINNsFormer architecture uses the following core components to enforce causal constraints across the temporal domain :

- **Pseudo-Sequence Generation**: For each temporal coordinate t, a short forward-marching temporal sequence of length k is constructed using a fixed look-ahead step size $\Delta t$ :

$$t \xrightarrow{\text{Generator}} \mathbf{T} = \left\{ t_0, t_1, \dots, t_{k-1} \right\} \in \mathbb{R}^k, \quad \text{where } t_\gamma = t + \gamma \Delta t$$

The parameter k determines the size of the look-ahead window, while $\Delta t$ determines the temporal coupling resolution.

- **Positional Encodings**: Sinusoidal positional encodings are added to the projected embeddings to preserve the temporal ordering of the pseudo-sequence :
  $$\mathbf{P}_{(\gamma, 2j)} = \sin\left(\frac{\gamma}{10000^{2j/d_{\text{model}}}}\right), \quad \mathbf{P}_{(\gamma, 2j+1)} = \cos\left(\frac{\gamma}{10000^{2j/d_{\text{model}}}}\right)$$
- **Spatio-Temporal Mixer**: A linear input layer projects the coordinates into a high-dimensional latent space:
  $$\mathbf{H}^{(0)} = \mathbf{T}\mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}} + \mathbf{P} \in \mathbb{R}^{k \times d_{\text{model}}}$$
- **Transformer Encoder-Decoder Layers**: Multi-head self-attention computes pairwise interactions across the temporal sequence, which propagates physical information along the temporal manifold and mimics the behavior of implicit time integration :
  $$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}$$
- **Wavelet Activation Function**: Standard activations (such as tanh or Swish) are replaced with a learnable Wavelet activation function :

$$\text{Wavelet}(z) = \omega_1 \sin(z) + \omega_2 \cos(z)$$

where $\omega_1, \omega_2$ are trainable, neuron-specific coefficients. This acts as a localized Fourier mode, enabling the network to resolve high-frequency components of the upstroke and overcome spectral bias.

- **Sequential Loss**: The pointwise loss is replaced with a sequential loss that enforces physical constraints consistently across the entire look-ahead temporal sequence, serving as an implicit time-marching regularizer :

$$\mathcal{L}_{\text{seq\_residual}}(\theta) = \frac{1}{k N_r} \sum_{i=1}^{N_r} \sum_{\gamma=0}^{k-1} \left\| \mathcal{D}\left - \mathbf{f}(t_i + \gamma \Delta t) \right\|^2$$

Designing a PINNsFormer sequence representation for the 4D Hodgkin-Huxley state $\mathbf{u}(t) = [V, m, h, n]^T$ requires evaluating three sequence design paradigms:

| Paradigm | Design Structure | Mathematical Formulation | Advantages | Disadvantages |
|---|---|---|---|---|
| **I: Sliding Time Windows** | Single temporal coordinate is expanded into a sequence of time steps for all 4 variables. | $\mathbf{H}^{(0)} \in \mathbb{R}^{k \times d_{\text{model}}}$, Output $\hat{\mathbf{u}} \in \mathbb{R}^{k \times 4}$. | Aligns with the standard PINNsFormer formulation; enforces strong temporal causality. | Cannot model independent timescales for different state variables. |
| **II: State Trajectories** | State variables are treated as separate tokens in the sequence, decoupling time and variables. | $\mathbf{H}^{(0)} \in \mathbb{R}^{4 \times d_{\text{model}}}$, representing $\{V, m, h, n\}$. | Directly models coupling and interactions between variables. | Lacks explicit temporal look-ahead modeling within the tokens. |
| **III: Joint Temporal-State Embedding** | Generates a grid sequence representing both temporal steps and individual variables. | $\mathbf{H}^{(0)} \in \mathbb{R}^{4k \times d_{\text{model}}}$, projecting state-time coordinates. | Captures both temporal marching and cross-variable kinetics. | High memory overhead; attention matrix scales quadratically to $\mathcal{O}(16k^2)$. |

# Part 7: Architectural Variants for the Hodgkin-Huxley Dynamics

## Variant A: Single-Sequence Multi-Variable PINNsFormer

Variant A projects the temporal coordinate into a single look-ahead sequence of length k, where each token represents the joint 4D biophysical state of the neuron.

```
 +-----------------------+
 |  Temporal Coord (t)   |
 +-----------+-----------+
             |
+------------v-----------+
```

```
                 | Pseudo-Sequence Gen (k) |
                 +------------+------------+
                              |
                 +------------v------------+
                 | Spatio-Temporal Mixer   |
                 +------------+------------+
                              |
                 +------------v------------+
                 |   Transformer Encoder   |
                 +------------+------------+
                              |
                 +------------v------------+
                 |   Transformer Decoder   |
                 +------------+------------+
                              |
                 +------------v------------+
                 |  Wavelet Activation     |
                 +------------+------------+
                              |
                 +------------v------------+
                 |  Multi-Variable Project |
                 +------------+------------+
                              |
      +-----------------------+----------------------+
      |                       |                      |
      v                       v                      v
Predicted V             Predicted m            Predicted h/n
{\hat{V}_0...\hat{V}_k} {\hat{m}_0...\hat{m}_k}
{\hat{h}_0/\hat{n}_0...}
```

## Mathematical Formulation

Given an input temporal coordinate t :

\mathbf{T} = \left^T \in \mathbb{R}^{k \times 1} \mathbf{H}^{(0)} = \text{Wavelet}(\mathbf{T}\mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}}) + \mathbf{P} \in \mathbb{R}^{k \times d_{\text{model}}} \mathbf{H}^{(l)} = \text{LN}\left(\mathbf{H}^{(l-1)} + \text{MHA}_{\text{self}}\left(\mathbf{H}^{(l-1)}\right)\right), \quad l = 1, \dots, L_{\text{enc}} \mathbf{D}^{(l)} = \text{LN}\left(\mathbf{D}^{(l-1)} + \text{MHA}_{\text{cross}}\left(\mathbf{D}^{(l-1)}, \mathbf{H}^{(L_{\text{enc}})}, \mathbf{H}^{(L_{\text{enc}})}\right)\right), \quad l = 1, \dots, L_{\text{dec}} \hat{\mathbf{U}} = \mathbf{D}^{(L_{\text{dec}})}\mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}} \in \mathbb{R}^{k \times 4} \hat{\mathbf{u}}(t + \gamma \Delta t) = \hat{\mathbf{U}}_{\gamma, \cdot} = \left[ \hat{V}_{\gamma}, \hat{m}_{\gamma}, \hat{h}_{\gamma}, \hat{n}_{\gamma} \right]^T

## Evaluation

- **Advantages**: Highly stable; simple parameter mapping.

- **Weaknesses**: Prone to capacity limitations during rapid upstrokes due to uniform temporal grouping.
- **Computational Complexity**: $\mathcal{O}(k \cdot d_{\text{model}} + k^2 \cdot d_{\text{model}})$ per forward pass.

## Variant B: Multivariate State-Token PINNsFormer

Variant B represents each physical state variable as an independent channel token, using self-attention to capture the biophysical coupling between variables.

```
                  +------------------------+
                  |  Temporal Coord (t)    |
                  +-----------+------------+
                              |
                 +------------v------------+
                 | Pseudo-Sequence Gen (k) |
                 +------------+------------+
                              |
              +---------------+---------------+
              |                               |
              v                               v
      [Mixer Project 1]               [Mixer Project 2]
              |                               |
              v                               v
       State-Token V                   State-Token m/h/n
  (k x d_model sequence)            (k x d_model sequence)
              |                               |
              +---------------+---------------+
                              |
                              v
              +-------------------------------+
              |  Channel Concatenation Layer  |
              |   (4k x d_model sequence)     |
              +---------------+---------------+
                              |
                              v
              +-------------------------------+
              |   Transformer Encoder Stack   |
              | (Inter-Channel Cross Attention)|
              +---------------+---------------+
                              |
                              v
              +-------------------------------+
              |   State-Specific Output Head  |
              +-------------------------------+
```

### Mathematical Formulation

The sequence generation is split into coordinate-specific representations:

$$\mathbf{T}_{\gamma} = t + \gamma \Delta t, \quad \gamma \in \{0, \dots, k-1\}$$

$$\mathbf{E}_{\gamma}^{V} = \text{Wavelet}(\mathbf{T}_{\gamma}\mathbf{W}_{\text{in}}^{V} + \mathbf{b}_{\text{in}}^{V}) + \mathbf{C}_V \in \mathbb{R}^{1 \times d_{\text{model}}}$$

$$\mathbf{E}_{\gamma}^{w} = \text{Wavelet}(\mathbf{T}_{\gamma}\mathbf{W}_{\text{in}}^{w} + \mathbf{b}_{\text{in}}^{w}) + \mathbf{C}_w \in \mathbb{R}^{1 \times d_{\text{model}}}, \quad w \in \{m, h, n\}$$

where $\mathbf{C}_i$ are learned channel-identity embeddings. These representations are concatenated to form the input sequence:

$$\mathbf{H}^{(0)} = \left[ \mathbf{E}_0^V, \dots, \mathbf{E}_{k-1}^V, \mathbf{E}_0^m, \dots, \mathbf{E}_{k-1}^m, \mathbf{E}_0^h, \dots, \mathbf{E}_{k-1}^h, \mathbf{E}_0^n, \dots, \mathbf{E}_{k-1}^n \right]^T \in \mathbb{R}^{4k \times d_{\text{model}}}$$

$$\mathbf{H}^{(L)} = \text{TransformerEncoder}\left(\mathbf{H}^{(0)}\right) \in \mathbb{R}^{4k \times d_{\text{model}}}$$

Each variable slice is projected independently to produce its look-ahead sequence:

$$\hat{\mathbf{V}} = \mathbf{H}_{[0:k], \cdot}^{(L)}\mathbf{w}_{\text{out}}^{V} + b_{\text{out}}^{V} \in \mathbb{R}^{k \times 1}$$

$$\hat{\mathbf{w}} = \mathbf{H}_{[\star:\star], \cdot}^{(L)}\mathbf{w}_{\text{out}}^{w} + b_{\text{out}}^{w} \in \mathbb{R}^{k \times 1}, \quad w \in \{m, h, n\}$$

### Evaluation

- **Advantages**: Learns non-linear interactions and feedback paths directly within the attention weights.
- **Weaknesses**: Prone to high memory overhead due to the 4k sequence length.
- **Computational Complexity**: $\mathcal{O}(16k^2 \cdot d_{\text{model}})$ per forward pass.

## Variant C: Cross-Attention Gating PINNsFormer

Variant C decouples the architecture into two sub-networks: a fast-state encoder that models the fast variables (V, m) and a slow-state decoder that models the slow variables (h, n), using cross-attention to coordinate their interaction.

```
+--------------------------------------------------------+
|                 Temporal Coordinate (t)                |
+-----------+-----------------------------------+--------+
            |                                   |
            v                                   v
+-----------------------+           +-----------------------+
| Fast Pseudo-Sequence  |           | Slow Pseudo-Sequence  |
|     Gen (k_fast)      |           |     Gen (k_slow)      |
+-----------+-----------+           +-----------+-----------+
            |                                   |
            v                                   v
+-----------------------+           +-----------------------+
|   Fast state Mixer    |           |   Slow state Mixer    |
+-----------+-----------+           +-----------+-----------+
            |                                   |
            v                                   v
+-----------------------+                       |
```

```
| Fast Transformer Enc  |                         |
|     (V, m dynamics)   |                         |
+-----------+-----------+                         |
            |                                     |
            +-----------------+                   |
                              |                   |
                              v                   v
                        +-----x-------------------+-------+
                        |     Cross-Attention Decoder     |
                        |  (Fuses Fast & Slow kinetics)   |
                        +---------------+-----------------+
                                        |
                                        v
                        +---------------+-----------------+
                        |   Joint Decoupled Output Head   |
                        +---------------------------------+
```

### Mathematical Formulation

The model uses decoupled coordinate sets to represent the fast and slow dynamics:

$$\mathbf{T}_{\text{fast}} = \left^T \in \mathbb{R}^{k_f \times 1}$$
$$\mathbf{T}_{\text{slow}} = \left^T \in \mathbb{R}^{k_s \times 1}, \quad \text{where } \Delta t_s \gg \Delta t_f$$
$$\mathbf{H}_{\text{fast}} = \text{TransformerEncoder}\left(\mathbf{T}_{\text{fast}}\mathbf{W}_{\text{in}}^{f} + \mathbf{b}_{\text{in}}^{f}\right) \in \mathbb{R}^{k_f \times d_{\text{model}}}$$
$$\mathbf{D}_{\text{slow}}^{(0)} = \mathbf{T}_{\text{slow}}\mathbf{W}_{\text{in}}^{s} + \mathbf{b}_{\text{in}}^{s} \in \mathbb{R}^{k_s \times d_{\text{model}}}$$
$$\mathbf{D}_{\text{slow}}^{(l)} = \text{LN}\left(\mathbf{D}_{\text{slow}}^{(l-1)} + \text{MHA}_{\text{cross}}\left(\mathbf{D}_{\text{slow}}^{(l-1)}, \mathbf{H}_{\text{fast}}, \mathbf{H}_{\text{fast}}\right)\right), \quad l = 1, \dots, L$$
$$\hat{\mathbf{u}}_{\text{slow}} = \mathbf{D}_{\text{slow}}^{(L)}\mathbf{W}_{\text{out}}^{s} + \mathbf{b}_{\text{out}}^{s} \in \mathbb{R}^{k_s \times 2}$$

### Evaluation

- **Advantages**: Biophysically aligned with the fast-slow decomposition, which reduces stiffness-induced optimization instabilities.
- **Weaknesses**: Requires tuning separate sequence lengths $k_f$, $k_s$.
- **Computational Complexity**: $\mathcal{O}(k_f^2 \cdot d_{\text{model}} + k_f k_s \cdot d_{\text{model}})$ per forward pass.

## Variant D: Hierarchical Timescale PINNsFormer

Variant D utilizes a multi-rate sequence generator to capture different timescales, using a fine temporal resolution for the fast variables (V, m) and a coarse, long-horizon resolution for the slow variables (h, n).

```
+----------------------+
|  Temporal Coord (t)   |
```

```
                     +-----------+-----------+
                                 |
                 +---------------+---------------+
                 |                               |
                 v
[span_108](start_span)[span_108](end_span)[span_111](start_span)[span_
111](end_span)v
        +-----------------------+       +-----------------------+
        |   Fine Sequence Gen   |       |   Coarse Sequence Gen |
        |    (k_f, dt_fine)     |       |    (k_c, dt_coarse)   |
        +-----------+-----------+       +-----------+-----------+
                    |                               |
                    v                               v
        +-----------------------+       +-----------------------+
        |   Fast state Mixer    |       |    Slow state Mixer   |
        +-----------+-----------+       +-----------+-----------+
                    |                               |
                    v                               v
        +-----------------------+       +-----------------------+
        |  Fast Transformer Enc |       |  Slow Transformer Enc |
        +-----------+-----------+       +-----------+-----------+
                    |                               |
                    +---------------+---------------+
                                    |
                                    v
        +---------------------------x---------------------------+
        |                Temporal Multiscale Alignment          |
        |                 (Linear Interpolation / MHA)          |
        +---------------------------+---------------------------+
                                    |
                                    v
        +---------------------------v---------------------------+
        |               State Output Projection Head            |
        +-------------------------------------------------------+
```

### Mathematical Formulation

The temporal coordinates are generated on two distinct scales:
$\mathbf{T}_{\text{fine}} = \left^T \in \mathbb{R}^{k_f \times 1} \mathbf{T}_{\text{coarse}} = \left^T \in \mathbb{R}^{k_c \times 1}, \quad \text{where } \Delta t_c \gg \Delta t_f$
These coordinates are processed through independent encoder paths:
$\mathbf{H}_{\text{fine}} = \text{TransformerEncoder}_{\text{fine}}\left(\mathbf{T}_{\text{fine}}\mathbf{W}_{\text{in}}^{f}\right) \in \mathbb{R}^{k_f \times d_{\text{model}}} \mathbf{H}_{\text{coarse}} = \text{TransformerEncoder}_{\text{coarse}}\left(\mathbf{T}_{\text{coarse}}\mathbf{W}_{\text{in}}^{c}\right) \in \mathbb{R}^{k_c \times d_{\text{model}}}$
These representations are aligned using a linear interpolation operator, $\mathbf{\Phi} \in$

$\mathbb{R}^{k_f \times k_c}$:

$$\mathbf{H}_{\text{coarse}}^{\text{aligned}} = \mathbf{\Phi} \mathbf{H}_{\text{coarse}} \in \mathbb{R}^{k_f \times d_{\text{model}}} \hat{\mathbf{U}} = \text{MLP}\left( \left[ \mathbf{H}_{\text{fine}} \, \Vert \, \mathbf{H}_{\text{coarse}}^{\text{aligned}} \right] \right) \in \mathbb{R}^{k_f \times 4}$$

**Evaluation**

- **Advantages**: Resolves both rapid action potential upstrokes and slow post-spike decay phases.
- **Weaknesses**: Prone to interpolation errors at transition boundaries.
- **Computational Complexity**: $\mathcal{O}\left((k_f^2 + k_c^2)\cdot d_{\text{model}}\right)$ per forward pass.

## Variant E: Neural Operator + PINNsFormer Hybrid

Variant E combines the spatial generalization of Neural Operators with the sequence-dependent temporal modeling of PINNsFormer. It uses a Branch net to encode external stimulus current profiles, $I_{\text{ext}}(t)$, and a Trunk net based on the PINNsFormer backbone.

```
+----------------------------+
+----------------------------+
|   Current Stimulus I(t)    |                |   Temporal Coord (t)
|
+-------------+-------------+
+-------------+-------------+
              |                                           |
              v                                           v
+-------------v-------------+
+-------------v-------------+
|    Neural Operator Branch |                |    PINNsFormer Trunk
|
|    (MLP / FNO Embedding)  |                |  (Look-Ahead Sequence
Enc)|
+-------------+-------------+
+-------------+-------------+
              |                                           |
              |              +---------------------------+
              |              |
              v              v
+-------------v--------------v-------------+
|            Hadamard Product              |
|            & Output Project              |
+--------------------+--------------------+
                     |
                     v
+--------------------+--------------------+
|        Predicted State Trajectory        |
|              \hat{u}(t)                  |
```

+ - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - +

**Mathematical Formulation**

Let the input function space represent the external current $I_{\text{ext}}(t)$ sampled at discrete points: $\mathbf{a} = [I(t_1), \dots, I(t_M)]^T \in \mathbb{R}^M$. The branch network projects this input into a latent feature space:

$$\mathbf{B}(\mathbf{a}) = \mathbf{W}_b^{(2)}\text{Wavelet}\left(\mathbf{W}_b^{(1)}\mathbf{a} + \mathbf{b}_b^{(1)}\right) + \mathbf{b}_b^{(2)} \in \mathbb{R}^{p \times d_{\text{model}}}$$

The Trunk net is parameterized as a PINNsFormer architecture that processes the temporal coordinates:

$$\mathbf{F}(t) = \text{TransformerEncoder}\left(\mathbf{T}\mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}}\right) \in \mathbb{R}^{k \times d_{\text{model}}}$$

The physical prediction at step $\gamma$ is computed using a multi-head outer product :

$$\hat{u}_i(t[span_176](start_span)[span_176](end_span)_{\gamma}; \mathbf{a}) = \sum_{j=1}^{d_{\text{model}}} \mathbf{B}_{i, j}(\mathbf{a}) \cdot \mathbf{F}_{\gamma, j}(t) + b_i$$

**Evaluation**

- **Advantages**: Generalizes across diverse, unseen stimulus profiles without retraining.
- **Weaknesses**: Prone to optimization instabilities during joint training of the Branch and Trunk networks.
- **Computational Complexity**: $\mathcal{O}(M \cdot d_{\text{model}} + k^2 \cdot d_{\text{model}})$ per forward pass.

# Part 8: Attention Map Interpretation and Biophysical Alignment

The multi-head self-attention matrix computed in PINNsFormer operates as a coordinate-coupling operator. For a single attention head, the attention weight $\mathbf{A}_{\gamma, \gamma'}$ measures the correlation between hidden states at step $\gamma$ and step $\gamma'$ of the look-ahead sequence :

$$\mathbf{A}_{\gamma, \gamma'} = \frac{\exp\left(\frac{\mathbf{q}_{\gamma}\mathbf{k}_{\gamma'}^T}{\sqrt{d_k}}\right)}{\sum_{j=1}^k \exp\left(\frac{\mathbf{q}_{\gamma}\mathbf{k}_{j}^T}{\sqrt{d_k}}\right)}$$

where $\mathbf{q}$ and $\mathbf{k}$ are query and key projections of the latent coordinates.
During training on the 4D Hodgkin-Huxley model, the self-attention matrices learn to adapt to the physical state of the system. This adaptive behavior is illustrated by the following attention maps (evaluated for a sequence length of $k=4$):

## Spike Onset Phase (Upstroke Peak)

During the rapid sodium activation phase, the attention map concentrates along the main diagonal :

| Query / Key | $\gamma = 0$ | $\gamma = 1$ | $\gamma = 2$ | $\gamma = 3$ |
|---|---|---|---|---|
| **$\gamma = 0$** | 0.90 | 0.10 | 0.00 | 0.00 |

| Query / Key | \gamma = 0 | \gamma = 1 | \gamma = 2 | \gamma = 3 |
|---|---|---|---|---|
| **\gamma = 1** | 0.05 | 0.85 | 0.10 | 0.00 |
| **\gamma = 2** | 0.00 | 0.05 | 0.90 | 0.05 |
| **\gamma = 3** | 0.00 | 0.00 | 0.10 | 0.90 |

*Biophysical Interpretation*: The attention focus allows the network to isolate the rapid upstroke dynamics, preventing the low-frequency spectral bias of standard MLPs from smoothing out the spike peak.

### Repolarization & Refractory Phase

As the slow potassium channels activate, the attention weights exhibit an asymmetric temporal drift :

| Query / Key | \gamma = 0 | \gamma = 1 | \gamma = 2 | \gamma = 3 |
|---|---|---|---|---|
| **\gamma = 0** | 0.40 | 0.40 | 0.10 | 0.10 |
| **\gamma = 1** | 0.10 | 0.50 | 0.30 | 0.10 |
| **\gamma = 2** | 0.05 | 0.15 | 0.60 | 0.20 |
| **\gamma = 3** | 0.00 | 0.05 | 0.25 | 0.70 |

*Biophysical Interpretation*: The model queries backward in time across the look-ahead window, capturing the causal dependence of repolarization on the preceding depolarizing upstroke.

### Steady-State Quiescence Phase

During the resting phase, the attention map becomes nearly uniform :

| Query / Key | \gamma = 0 | \gamma = 1 | \gamma = 2 | \gamma = 3 |
|---|---|---|---|---|
| **\gamma = 0** | 0.25 | 0.25 | 0.25 | 0.25 |
| **\gamma = 1** | 0.25 | 0.25 | 0.25 | 0.25 |
| **\gamma = 2** | 0.25 | 0.25 | 0.25 | 0.25 |
| **\gamma = 3** | 0.25 | 0.25 | 0.25 | 0.25 |

*Biophysical Interpretation*: The flat attention profile allows the network to aggregate low-frequency information, stabilizing the predictions under the stable resting focus.
These attention maps have a clear biophysical interpretation. Rather than acting as arbitrary data-fitting structures, the attention weights recover the underlying state-dependent transition probabilities. When the system is in a highly stiff, fast-moving phase, the attention mechanism dynamically shrinks the temporal context window, acting like a self-adaptive variable time-step integrator. This dynamic routing propagates information through the fast-slow state manifold, helping the model resolve bifurcations and maintain physical causality.

# Part 9: Stiffness, Multiscale Dynamics, and Comparative Framework Analysis

Stiffness and multiscale timescale separation present major optimization bottlenecks for physics-informed machine learning architectures. In standard coordinate-based PINNs, gradients are evaluated pointwise, causing the loss landscape to become highly non-convex near steep gradients. Consequently, first-order gradient updates oscillate wildly across the steep valleys of fast transient modes, preventing the network from resolving slow modes and leading to optimization plateaus where the model fails to capture action potential spikes.

The sequential loss and coordinate-coupling attention in PINNsFormer mitigate these failure modes. By linking predictions across k temporal steps, the sequential loss acts as an implicit time-marching regularizer. This bounds the local Lipschitz constant of the gradient updates and smooths the loss landscape, preventing the optimizer from stalling.

The following table compares the performance of PINNsFormer against alternative stiffness mitigation and scientific machine learning strategies on the 4D Hodgkin-Huxley system :

| Framework | Optimization Strategy | Computational Complexity | Stability in Stiff Regimes | Vulnerability to Spectral Bias | Long-Horizon Performance | Required Data Volume |
|---|---|---|---|---|---|---|
| **Vanilla PINN** | Global pointwise residual minimization. | $\mathcal{O}(N_{\text{col}})$. | Very Low; fails during fast upstrokes. | Extremely High; over-smooths spikes. | Poor; suffers from temporal decay. | Zero (mesh-free forward solver). |
| **Stiff-PINN (STL-PINN)** | Stiff transfer learning using step-wise parameter scaling. | $\mathcal{O}(M \cdot N_{\text{col}})$. | High; tracks slow parameters sequentially. | Moderate; resolves high frequencies via transfer stages. | Moderate; requires sequential restarts. | Low; requires parameter-grid initialization. |
| **Adaptive PINN** | Dynamic residual loss weighting. | $\mathcal{O}(N_{\text{col}})$. | Moderate; balances gradient magnitudes. | High; still limited by the MLP backbone. | Poor; susceptible to temporal accumulation errors. | Zero (uses residual weights). |
| **XPINN** | Domain decomposition into parallel subdomains. | $\mathcal{O}(N_{\text{sub}} \cdot N_{\text{col}})$. | High; isolates stiff regions in space-time. | Moderate; resolves localized gradients. | High; requires interface alignment constraints. | Zero (mesh-free forward solver). |
| **DeepONet / FNO** | Operator learning via branch-trunk mappings. | $\mathcal{O}(N_{\text{params}})$. | High; bypasses stiff integration during inference. | Low; models continuous operators directly. | High; requires boundary alignment. | Extremely High; requires large, high-fidelity datasets. |
| **Neural ODE** | Explicit integration using adaptive numerical solvers. | $\mathcal{O}(N_{\text{steps}})$. | Extremely High; leverages stiff solvers (e.g., Radau). | Low; uses analytical integration paths. | Excellent; maintains physical causality. | High; requires continuous training trajectories. |
| **PINNsFormer** | Look-ahead sequence modeling with sequential loss. | $\mathcal{O}(k^2 \cdot N_{\text{col}})$. | Extremely High; attention smooths the loss landscape. | Low; Wavelet activation resolves high frequencies. | Excellent; enforces causality across the domain. | Zero (uses sequential physics loss). |

# Part 10: Exhaustive Numerical Experiments and Evaluation Roadmap

To validate the capabilities of PINNsFormer on the 4D Hodgkin-Huxley system, this section defines a series of rigorous numerical experiments.

```
+-----------------------------------+
|       Experimental Inputs         |
| (Stimulus I_ext, Noise, Params)   |
+-----------------+-----------------+
                  |
     [PINNsFormer Architecture]
                  |
                  v
+-----------------------------------+
|      Predicted Trajectories       |
|     \hat{\mathbf{u}}_{\theta}(t)  |
+-----------------+-----------------+
                  |
   +--------------+--------------+
   |                             |
   v                             v
+--------------------+       +--------------------+
|   State Residual   |       |    Parameter Loss  |
|      Loss (L_r)    |       |    (For Inverse)   |
+---------+----------+       +---------+----------+
          |                            |
          +--------------+-------------+
                         |
                         v
+-----------------------------------+
|  Metric Evaluation:               |
|  (rMAE, rRMSE, Phase-Divergence,  |
|   Spike-Timing Precision)         |
+-----------------------------------+
```

## Experiment 1: Single Spike Generation

- **Biophysical Configuration**: Membrane stimulated with a short current pulse $I_{\text{ext}} = 10.0\,\mu\text{A/cm}^2$ for $t \in [0, 2.0\text{ ms}]$, evaluated over a temporal domain of $t \in [0, 20.0\text{ ms}]$.
- **State Initializations**: Steady-state rest conditions: $V_0 = -65.0\text{ mV}$, $m_0 = 0.0529$, $h_0 = 0.5961$, $n_0 = 0.3177$.
- **Evaluation Metrics**:
    - Relative Root Mean Square Error of Voltage:

\text{rRMSE}_V = \sqrt{\frac{\sum_{i=1}^{N} (V(t_i) - \hat{V}_i)^2}{\sum_{i=1}^{N}

V(t_i)^2}}

- Absolute error in predicted peak voltage: $\Delta V_{\text{peak}} = |V_{\text{peak}} - \hat{V}_{\text{peak}}|$.
- **Expected Failure Modes**: The network may fail to resolve the action potential upstroke, smoothing the peak voltage and delaying the repolarization phase.

## Experiment 2: Repeated Firing (Tonic Spiking)

- **Biophysical Configuration**: Membrane stimulated with a sustained DC current $I_{\text{ext}} = 15.0\,\mu\text{A/cm}^2$ over a long temporal domain of $t \in [0, 100.0\text{ ms}]$, forcing the system past the Hopf bifurcation point into stable limit cycle oscillations.
- **State Initializations**: Steady-state rest conditions.
- **Evaluation Metrics**:
  - Relative error in predicted firing frequency:
  $\text{Error}_f = \frac{|f_{\text{true}} - f_{\text{pred}}|}{f_{\text{true}}}$
- Mean Absolute Error of Inter-Spike Intervals ($\text{ISI}$):
  $\text{MAE}_{\text{ISI}} = \frac{1}{M-1} \sum_{j=1}^{M-1} |\text{ISI}_j^{\text{true}} - \text{ISI}_j^{\text{pred}}|$
- **Expected Failure Modes**: Phase drift and accumulated temporal integration errors, where the predicted spikes drift out of phase relative to the true solution over long simulation horizons.

## Experiment 3: Bursting Dynamics

- **Biophysical Configuration**: Multiscale slow-harmonic stimulus $I_{\text{ext}}(t) = 5.0\sin(0.1 t) + 8.0\,\mu\text{A/cm}^2$ over $t \in [0, 500\text{ ms}]$, or parameters adjusted to simulate a bursting regime.
- **State Initializations**: Quiescent state boundaries.
- **Evaluation Metrics**:
  - Accuracy of the predicted active-phase duration ($\Delta T_{\text{active}}$) and silent-phase duration ($\Delta T_{\text{silent}}$).
  - Phase-Space Divergence ($\text{PSD}$) of the slow-manifold trajectories:
  $\text{PSD} = \frac{1}{N} \sum_{i=1}^N \sqrt{(h_i - \hat{h}_i)^2 + (n_i - \hat{n}_i)^2}$
- **Expected Failure Modes**: The model may fail to capture the transition boundaries, either predicting continuous spiking or failing to trigger the bursting phase due to timescale mismatch.

## Experiment 4: Biophysical Parameter Estimation

- **Biophysical Configuration**: Voltage observations $V_{\text{obs}}(t)$ recorded under a constant stimulus $I_{\text{ext}} = 10.0\,\mu\text{A/cm}^2$ for $t \in [0, 50\text{ ms}]$, with unknown conductances $\bar{g}_{\text{Na}}$, $\bar{g}_{\text{K}}$, $g_{\text{L}}$ treated as trainable model parameters.
- **State Initializations**: Gating variables initialized to flat, non-informative profiles ($m, h, n = 0.5$).
- **Evaluation Metrics**:
  - Parameter estimation error:
  $\text{Error}_{\theta} = \frac{|\theta_{\text{true}} - \theta_{\text{pred}}|}{\theta_{\text{true}}}$

$\times 100\%$
- Hidden state reconstruction error: $\text{rRMSE}_w$ for $w \in \{m, h, n\}$.
- **Expected Failure Modes**: Prone to local minima and parameter identifiability issues, where different combinations of conductances yield similar voltage profiles.

### Experiment 5: Parameter Recovery Under Measurement Noise

- **Biophysical Configuration**: Membrane potential observations corrupted by additive zero-mean Gaussian noise with variance scaled to simulate realistic SNR levels ($\text{SNR} \in \{10\text{ dB}, 20\text{ dB}, 30\text{ dB}\}$).
- **State Initializations**: Steady-state rest conditions.
- **Evaluation Metrics**:
  - Robustness parameter sensitivity curves plotting parameter estimation error as a function of SNR.
  - Filtering efficiency measured by the reconstruction error relative to the clean ground-truth.
- **Expected Failure Modes**: High-frequency measurement noise may degrade the automatic differentiation of residuals, causing the parameter estimates to diverge.

### Experiment 6: Long-Time Forecasting (Extrapolation)

- **Biophysical Configuration**: Model trained on a short temporal window $t \in [0, 20.0\text{ ms}]$ under constant stimulation, then evaluated on its ability to extrapolate trajectories over $t \in [20.0\text{ ms}, 100.0\text{ ms}]$.
- **State Initializations**: Steady-state rest conditions.
- **Evaluation Metrics**:
  - Relative error ($\text{rRMSE}$) as a function of the extrapolation distance.
  - Extrapolated phase-space trajectory divergence relative to the true limit cycle.
- **Expected Failure Modes**: Fast accumulation of phase errors, with the extrapolated trajectory collapsing back to a stable focus or drifting toward unphysical states.

### Experiment 7: Out-of-Distribution Stimulation Currents

- **Biophysical Configuration**: Model evaluated on complex, unseen current profiles (such as ramped or chaotic Lorenz-driven stimuli) after training on step currents.
- **State Initializations**: Steady-state rest conditions.
- **Evaluation Metrics**:
  - Mean squared error ($\text{MSE}$) across different stimulus profiles.
  - Qualitative assessment of predicted spike thresholds.
- **Expected Failure Modes**: Generalization failures if the network overfits the simple step profiles used during training.

# Part 11: Inverse Problems, Parameter Estimation, and Observability Analysis

In experimental computational neuroscience, we can typically only record the membrane potential $V(t)$, while the underlying gating variables $m(t)$, $h(t)$, $n(t)$ and the biophysical parameters \boldsymbol{\theta} remain unobserved.

```
  Observable
Mea[span_242](start_span)[span_242](end_span)[span_244](start_span)[span_244](end_span)surement: Voltage V(t)
           |
           v
  +-----------------------------------+
  |   Lie Derivative Observability    |
  |             Analysis              |
  +----------------+------------------+
                   |
                   v
  +-----------------------------------+
  |   Information Sensitivity (ISF)   |
  |             Analysis              |
  +----------------+------------------+
                   |
                   v
  +-----------------------------------+
  |     Parameter Identification      |
  |  (g_Na, g_K, g_L, E_Na, E_K, E_L) |
  +-----------------------------------+
```

To evaluate the mathematical feasibility of recovering the full state vector and parameters from voltage observations alone, we analyze the system's observability using **differential geometry and Lie derivatives**.

Let the nonlinear system be defined as :

\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \quad y = h(\mathbf{x}) = V = x_1

The Lie derivative of the scalar measurement function h(\mathbf{x}) along the vector field \mathbf{f}(\mathbf{x}) is defined recursively as :

L_{\mathbf{f}}^0 h(\mathbf{x}) = h(\mathbf{x}) = x_1 L_{\mathbf{f}}^1 h(\mathbf{x}) = \nabla\left(L_{\mathbf{f}}^0 h\right) \cdot \mathbf{f}(\mathbf{x}) = f_1(\mathbf{x}) = \frac{1}{C_m} \left( I_{\text{ext}} - \bar{g}_{\text{Na}}x_2^3x_3(x_1 - E_{\text{Na}}) - \bar{g}_{\text{K}}x_4^4(x_1 - E_{\text{K}}) - g_{\text{L}}(x_1 - E_{\text{L}}) \right) L_{\mathbf{f}}^2 h(\mathbf{x}) = \nabla\left(L_{\mathbf{f}}^1 h\right) \cdot \mathbf{f}(\mathbf{x}) = \frac{\partial f_1}{\partial x_1}f_1 + \frac{\partial f_1}{\partial x_2}f_2 + \frac{\partial f_1}{\partial x_3}f_3 + \frac{\partial f_1}{\partial x_4}f_4 L_{\mathbf{f}}^3 h(\mathbf{x}) = \nabla\left(L_{\mathbf{f}}^2 h\right) \cdot \mathbf{f}(\mathbf{x})

The nonlinear observability matrix \mathbf{O}_M(\mathbf{x}) is constructed from the gradients of these Lie derivatives :

\mathbf{O}_[span_251](start_span)[span_251](end_span)[span_253](start_span)[span_253](end_span)[span_255](start_span)[span_255](end_span)M(\mathbf{x}) = \begin{bmatrix} \nabla L_{\mathbf{f}}^0 h(\mathbf{x}) \\ \nabla L_{\mathbf{f}}^1 h(\mathbf{x}) \\ \nabla L_{\mathbf{f}}^2 h(\mathbf{x}) \\ \nabla L_{\mathbf{f}}^3 h(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ J_{11} & J_{12} & J_{13} & J_{14} \\ \frac{\partial (L_{\mathbf{f}}^2 h)}{\partial x_1} & \frac{\partial (L_{\mathbf{f}}^2 h)}{\partial x_2} & \frac{\partial (L_{\mathbf{f}}^2 h)}{\partial x_3} & \frac{\partial (L_{\mathbf{f}}^2 h)}{\partial x_4} \\ \frac{\partial (L_{\mathbf{f}}^3 h)}{\partial x_1} & \frac{\partial (L_{\mathbf{f}}^3 h)}{\partial x_2} & \frac{\partial (L_{\mathbf{f}}^3 h)}{\partial x_3} &

$$\frac{\partial (L_{\mathbf{f}}^3 h)}{\partial x_4} \end{bmatrix}$$

A nonlinear system is locally observable at a state $\mathbf{x}_0$ if the observability matrix has full row rank :

$$\text{rank}(\mathbf{O}_M(\mathbf{x}_0)) = n = 4 \iff \det(\mathbf{O}_M(\mathbf{x}_0)) \neq 0$$

For the 4D Hodgkin-Huxley model, the derivatives of the voltage equation with respect to the gating variables are non-zero across the physiological domain ($V \neq E_{\star}$) :

$$\frac{\partial f_1}{\partial m} \propto m^2 h (V - E_{\text{Na}}) \neq 0 \frac{\partial f_1}{\partial h} \propto m^3 (V - E_{\text{Na}}) \neq 0 \frac{\partial f_1}{\partial n} \propto n^3 (V - E_{\text{K}}) \neq 0$$

Because these coupling terms are non-zero, the rows of $\mathbf{O}_M(\mathbf{x})$ are linearly independent. This confirms that the unobserved gating variables ($m$, $h$, $n$) and the biophysical parameters $\boldsymbol{\theta}$ are mathematically observable from voltage recordings alone, provided the voltage is not clamped to a constant value ($V \neq \text{constant}$).

To assess parameter identifiability during different phases of the action potential, we define **Information Sensitivity Functions (ISFs)**, which measure how the system state responds to perturbations in the parameters :

$$\mathbf{S}_{\theta}(t) = \frac{\partial \mathbf{x}(t)}{\partial \theta}$$

These sensitivity trajectories evolve according to the linearized system :

$$\frac{d\mathbf{S}_{\theta}}{dt} = \mathbf{J}(t)\mathbf{S}_{\theta} + \frac{\partial \mathbf{f}(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta}$$

Analyzing the sensitivity trajectories $\mathbf{S}_{\theta}(t)$ reveals that information about sodium conductances ($\bar{g}_{\text{Na}}$, $E_{\text{Na}}$) is concentrated within a narrow window near the action potential upstroke, whereas potassium and leak conductances ($\bar{g}_{\text{K}}$, $E_{\text{K}}$, $g_{\text{L}}$) are informed by the repolarization and hyperpolarization phases. PINNsFormer exploits these sensitivity profiles during parameter identification. It treats the inverse problem as a global optimization task over a temporal look-ahead window, utilizing self-attention to regularize parameter updates and ensure convergence to the true physical values.

# Part 12: Research Gaps, Proposal, and Promising PhD Thesis Directions

The application of scientific machine learning, particularly Transformer-based frameworks, to computational neuroscience is an emerging field. While PINNsFormer has demonstrated success on benchmark physical systems, several open research questions remain. This section ranks the top 20 open research questions, outlines a formal PhD research proposal, and suggests promising thesis directions.

## Ranked Open Research Questions in SciML for Neuroscience

| Rank | Research Question | Principal Technical Bottleneck | Potential Scientific Impact |
| --- | --- | --- | --- |
| **1** | Can we adapt linear attention mechanisms (such as Mamba or FlashAttention) to | Quadratic memory scaling of self-attention. | Enables simulation of large-scale cortical networks. |

| Rank | Research Question | Principal Technical Bottleneck | Potential Scientific Impact |
| --- | --- | --- | --- |
| | maintain temporal coupling while achieving $\mathcal{O}(k)$ complexity for long-horizon neural simulations? | | |
| **2** | How can we automate the selection of optimal spatial bases in Neuro-Spectral Architectures (NeuSA) to solve parameterized spatial Hodgkin-Huxley equations? | Spatial basis projection overhead. | Accurate propagation modeling in myelinated vs. unmyelinated axons. |
| **3** | How can we design coordinate-relative attention biases (similar to ALiBi) that respect irregular non-Euclidean dendritic geometries? | Mapping coordinate-relative biases to irregular topologies. | Resolves synaptic integration dynamics in complex reconstructed neurons. |
| **4** | What are the analytical error-propagation bounds of autoregressive PINNsFormer models over long time horizons? | Non-linear error accumulation across rollout steps. | Guarantees numerical stability for long-time neural forecasting. |
| **5** | Can we implement a learnable pseudo-step size $\Delta t$ within the generator that dynamically adjusts based on the local stiffness of the system? | Instability in the AD computational graph. | Resolves spike-induced stiffness with high computational efficiency. |
| **6** | How can we use attention matrix entropy to dynamically scale physics loss terms ($\lambda_r$, $\lambda_{\text{ic}}$), replacing empirical tuning schemes? | Extreme non-convex loss landscapes. | Eliminates optimization failures and speeds up convergence. |
| **7** | Can we construct sparse attention | Dynamic sparsity masking inside the | Reduces the computational footprint |

| Rank | Research Question | Principal Technical Bottleneck | Potential Scientific Impact |
|---|---|---|---|
| | patterns that track action potential upstrokes, concentrating resources on high-stiffness regions? | Transformer layer. | during non-spiking phases. |
| **8** | How can we train decoder-only PINNsFormer models to generalize to unseen boundary geometries without retraining? | Out-of-distribution generalization constraints. | Zero-shot inference for variable axon lengths and diameters. |
| **9** | Can we solve stochastic Hodgkin-Huxley equations (where ion channels exhibit random fluctuations) using generative Transformer backbones? | Infinite-dimensional noise-space integration. | Captures microscopic, channel-noise-driven spike variability. |
| **10** | Can we prove that the self-attention operator converges to the continuous Green's function of a given neural PDE family in the infinite-width limit? | Complexity of continuous operator representation. | Establishes a rigorous mathematical foundation for scientific Transformers. |
| **11** | How can we scale PINNsFormer to model large populations of coupled neurons (N > 10^4) with variable synaptic connection delays? | Quadratic scaling of multi-head attention. | Accelerates large-scale brain network modeling. |
| **12** | Can we use PINNsFormer to discover hidden biophysical variables from multi-electrode array (MEA) extracellular recordings? | Extreme partial observability and volume conduction noise. | Non-invasive reconstruction of localized neural circuits. |
| **13** | How can we embed energy-conservation | Modeling active thermodynamic | Ensures biophysically realistic metabolic |

| Rank | Research Question | Principal Technical Bottleneck | Potential Scientific Impact |
| --- | --- | --- | --- |
| | laws (like ATP consumption limits) directly into the attention weights of neural PINNs? | systems in PINN loss. | behavior. |
| **14** | Can we combine neural operator frameworks (like FNO) with PINNsFormer to solve parameterized neural field equations over cortical manifolds? | Spatial operator complexity on curved surfaces. | Bridges single-neuron dynamics with macro-scale EEG/MEG signals. |
| **15** | How can we implement hard initial conditions (HC-PINN) for coupled, high-dimensional neural ODE networks to remove gradient imbalances? | Highly coupled boundary constraints. | Eliminates the need to balance multi-objective loss terms. |
| **16** | Can we use PINNsFormer to model homeostatic plasticity, where conductances slowly adapt over hours or days? | Managing massive, multi-timescale separation ([span_180](start_span)[span_180](end_span) 10^{-3}\text{ s} to 10^5\text{ s}). | Resolves the biophysics of learning and memory. |
| **17** | How can we adapt PINNsFormer to solve fractional-order Hodgkin-Huxley equations, which capture memory effects in cell membranes? | High computational cost of fractional derivatives. | Models anomalous diffusion and pathological dynamics. |
| **18** | Can we integrate PINNsFormer with symbolic regression tools to discover new biophysical channel models from experimental patch-clamp data? | Discrete optimization over symbolic grammar spaces. | Replaces empirical curve-fitting with automated model discovery. |
| **19** | How can we train PINNsFormer models using mixed-precision or quantized weights to | Optimization failures under reduced precision. | Enables real-time, closed-loop brain-machine interfaces. |

| Rank | Research Question | Principal Technical Bottleneck | Potential Scientific Impact |
|---|---|---|---|
| | deploy them on low-power neuromorphic hardware? | | |
| **20** | Can PINNsFormer identify cellular-level pathological changes (such as demyelination or channelopathies) from macro-scale EEG waveforms? | Ill-posed inverse scaling across scales. | Translates computational neuroscience models into diagnostic tools. |

## Publication-Quality PhD Research Proposal

### Project Title

Multi-timescale Transformer-based Physics-Informed Neural Networks for High-Dimensional Neuronal Systems and Parameter Identification

### Abstract

Biophysically detailed neuron models, such as the 4D Hodgkin-Huxley system, are challenging to simulate and parameterize using standard numerical approaches due to extreme non-linearities, dynamic stiffness, and partial observability. This proposal introduces a multi-timescale PINNsFormer architecture designed to solve high-dimensional neuronal dynamics and infer biophysical parameters from sparse, noisy voltage measurements. By combining multi-head attention with a hierarchical sequence generator, the proposed model learns to capture fast upstrokes and slow decay phases across different timescales, mitigating the stiffness-related failures of classical PINNs.

```
                              |
      +-----------------------+-----------------------+
      |                                               |
      v                                               v

- Implement Hierarchical Architecture          - Scale to 2D Spatial
Dendrites
- Validate on Stiff Benchmarks                 - Implement Gated
Cross-Attention
- Evaluate Optimization Curricula              - Evaluate
Multi-Variable Networks
      |
[span_151](start_span)[span_151](end_span)          |
      +-----------------------+-----------------------+
                              |
                              v
```

```
[Months 25-36: Verification]
- Recover hidden gating variables
- Infer ATP metabolic constraints
- Deploy on neuromorphic hardware
```

### Specific Aims

- **Aim 1: Develop a Hierarchical, Multi-Timescale PINNsFormer Framework**: Design a sequence-to-sequence Transformer architecture with a multi-rate sequence generator to capture the disparate timescales of fast membrane potential upstrokes and slow channel inactivation.
- **Aim 2: Implement Real-Time Parameter Identification and Hidden State Reconstruction**: Evaluate the framework's ability to infer ionic conductances (\bar{g}_{\text{Na}}, \bar{g}_{\text{K}}, g_{\text{L}}) and reconstruct unobserved gating variables (m, h, n) from sparse, noisy voltage recordings.
- **Aim 3: Generalize the Model to Spatial Dendritic Geometry and Synaptic Coupling**: Extend the architecture to solve spatial Hodgkin-Huxley cable partial differential equations and model synaptic interactions in multi-neuron networks.

### Detailed Implementation Strategy

- **Architecture Design**: The model utilizes a Hierarchical Timescale PINNsFormer (Variant D) featuring dual sequence paths. A fine-timescale path (\Delta t = 0.05\text{ ms}) processes the fast variables (V, m), while a coarse-timescale path (\Delta t = 0.5\text{ ms}) processes the slow variables (h, n).
- **Optimization and Loss Formulation**: Training proceeds in two stages. Stage 1 uses a global search with the Adam optimizer over a sequential loss formulation to approximate the low-frequency physical manifold. Stage 2 utilizes the second-order L-BFGS optimizer to resolve high-frequency spike details with high precision.
- **Validation and Baseline Comparison**: Performance is benchmarked against standard PINNs, Stiff-PINNs (STL-PINN), and Neural ODEs across different stimulus regimes (tonic, bursting, and chaotic stimulation).

## Promising PhD Thesis Directions

### Direction 1: Closed-Loop Brain-Machine Interfaces via Neuromorphic PINNsFormers

- **Scientific Motivation**: Next-generation neural prosthetics require real-time, energy-efficient decoding and control of spiking neurons to interface with pathological brain regions.
- **Technical Approach**: This direction focuses on compressing and quantizing the PINNsFormer architecture into lightweight, low-rank matrices. By implementing hard-constraint initial conditions, the model can be deployed on neuromorphic hardware to perform real-time, closed-loop parameter estimation and neuromodulation.

### Direction 2: Neural Field Operators for Whole-Brain Dynamics and EEG

### Reconstruction

- **Scientific Motivation**: Understanding macro-scale brain activity requires models that bridge microscopic single-neuron biophysics with macro-scale electromagnetic signals (EEG/MEG).
- **Technical Approach**: This direction proposes a hybrid Neural Operator-Transformer (Variant E) to solve high-dimensional neural field partial differential equations over curved cortical manifolds. The framework learns continuous operator mappings between localized synaptic inputs and global cortical potentials, providing a biophysically consistent model for reconstructing EEG signals.

### Direction 3: Biophysical Model Discovery Under Metabolic and Thermodynamic Constraints

- **Scientific Motivation**: Empirical, curve-fitted channel representations often omit thermodynamic properties, such as ATP consumption and temperature-dependent gating kinetics.
- **Technical Approach**: This direction integrates symbolic regression with the learnable Wavelet activations of PINNsFormer to discover novel ion channel formulations directly from experimental patch-clamp data. The search space is constrained by imposing thermodynamic boundaries (such as the Goldman-Hodgkin-Katz current equation and metabolic energy limits), ensuring the discovered equations are biophysically realistic.

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